



FAIR 3D HERITAGE

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of the Digital 3D Heritage on the Web*

A Roadmap for Education-Driven Heritage Sustainability: The Case of Harput Castle

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Abstract

While 3D digital heritage technologies have significantly advanced the documentation and visualization of cultural assets through photogrammetry, laser scanning, and AI-assisted modeling, their broader societal impact—particularly in fostering long-term public engagement and intergenerational value transmission—remains largely unrealized. This study addresses that gap by proposing an education-driven sustainability roadmap for Harput Castle (Elazığ, Türkiye), a multi-layered archaeological site encompassing Hittite, Urartian, Roman, Byzantine, Seljuk, and Ottoman strata, with strong potential for UNESCO World Heritage List inscription. The proposed framework repositions digital twin systems, augmented reality (AR), virtual reality (VR), and multisensory experience technologies not as ends in themselves, but as enablers of a holistic educational engagement model centered on child-centered learning. Grounded in the recognition that older generations often maintain an inherent cultural connection to heritage, while younger generations require interactive, technology-mediated experiences to develop meaningful bonds, the framework introduces a “Cultural Ambassadors Program” that operates across three interconnected environments: on-site digital interactions where children explore AR-reconstructed historical layers; school-based curriculum-integrated activities including project-based learning and digital modeling; and home-based continuation kits that extend the heritage experience into the family setting, fostering intergenerational dialogue. Together, these components form a continuous engagement loop designed to transform children from passive visitors into active cultural custodians. As the project is currently at a conceptual stage, this contribution seeks to engage with the FAIR 3D Heritage community to learn from best practices in participatory design, education, and outreach, and to refine the proposed roadmap accordingly. By framing 3D heritage and digital twin systems as enablers of education-driven sustainability rather than solely visualization tools, the study offers a scalable and transferable model for strengthening public engagement and long-term cultural impact across heritage sites with similar potential.